Problem Set 68

The Mean Value Theorem

April 22, 2026

Problem 1

Verify that the function satisfies the three hypotheses of Rolle's Theorem on the given interval. Then find all numbers that satisfy the conclusion of Rolle's Theorem.

(b) $f(x) = \sqrt{x} - \dfrac{x}{3}, \quad [0, 9]$

1. $f(x)$ is continuous on $[0, 9]$ as $f(x)$ is constructed by polynomial functions and $x \geq 0$.
2. $f(x)$ is differentiable on $(0, 9)$.
3. $\begin{cases} f(0) = \sqrt{0} - \dfrac{0}{3} = 0 \\ f(9) = \sqrt{9} - \dfrac{9}{3} = 0 \end{cases} \implies f(0) = f(9)$

By Rolle's Theorem,

$$\boxed{\exists c \in (0, 9) \text{ s.t. } f'(c) = 0}$$

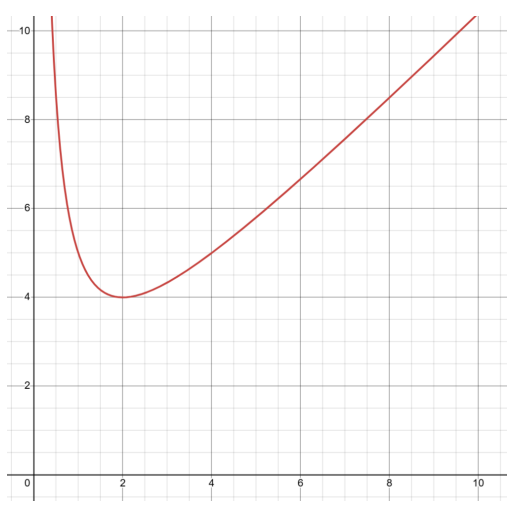
$$f'(c) = \frac{1}{2\sqrt{c}} - \frac{1}{3} = 0$$

$$2\sqrt{c} = 3$$

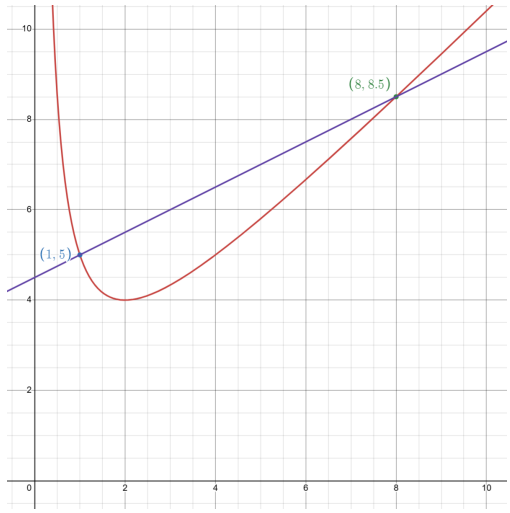
$$\boxed{c = \frac{9}{4}}$$

Problem 4

(a) Use Desmos to graph the function $f(x) = x + \dfrac{4}{x}$ in the viewing window $[0, 10]$ by $[0, 10]$.

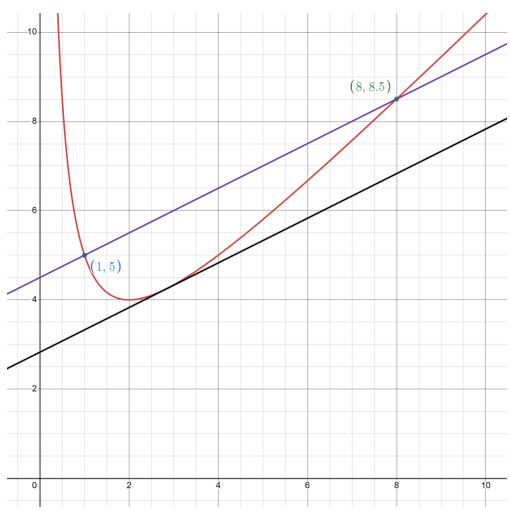


(b) Graph the secant line that passes through the points $(1, 5)$ and $(8, 8.5)$ on the same screen with f.



(c) Find the number c that satisfies the conclusion of the Mean Value Theorem for this function f and the interval $[1, 8]$. Then graph the tangent line at the point $(c, f(c))$ and notice that it is parallel to the secant line.

$$f'(c) = 1 - \frac{4}{c^2} = \frac{1}{2}$$
$$c^2 = 8$$
$$\boxed{c = 2\sqrt{2}}$$



Problem 5

Verify that the function satisfies the hypotheses of the Mean Value Theorem on the given interval. Then find all numbers c that satisfy the conclusion of the Mean Value Theorem.

(a) $f(x) = 3x^2 + 2x + 5, \quad [-1, 1]$

1. $f(x)$ is continuous on $[-1, 1]$ as $f(x)$ is constructed by polynomial functions.
2. $f(x)$ is differentiable on $(-1, 1)$.

By the Mean Value Theorem,

$$\boxed{\exists c \in (-1, 1) \text{ s.t. } f'(c) = 2}$$
$$f'(c) = 6c + 2 = 2$$
$$\boxed{c = 0}$$

Problem 6

Let $f(x) = (x-3)^{-2}$. Show that there is no value of c in $(1, 4)$ such that $f(4) - f(1) = f'(c)(4-1)$. Why does this not contradict the Mean Value Theorem?

$$f'(x) = -\frac{2}{(x-3)^3}$$
$$f(4) = 1, f(1) = \frac{1}{4}$$
$$f'(c) = -\frac{2}{(c-3)^3} = \frac{1}{4}$$
$$c - 3 = -2$$
$$c = 1 \implies c \notin (1, 4)$$

As $x \neq 3$, $f(x)$ is not continuous on $[1, 4]$. Hence, the conditions for the Mean Value Theorem are not satisfied.

Problem 7

Show that the equation $1 + 2x + x^3 + 4x^5 = 0$ has exactly one real root.

Given that $f(x)$ is constructed by polynomial functions, $f(x)$ is continuous and differentiable on $\mathbb{R}$. Assume two distinct roots a and b satisfying $a < b$ exist. By Rolle's Theorem, there exists $c \in (a, b)$ such that $f'(c) = 0$.

$$f'(x) = 20x^4 + 3x^2 + 2$$
$$f'(c) > 0$$

which contradicts the assumption. Thus, $f(x)$ has exactly one real root.

Problem 9

If $f(1) = 10$ and $f'(x) \geq 2$ for $1 \leq x \leq 4$, how small can $f(4)$ possibly be?

$$f'(c) = \frac{f(4) - f(1)}{4 - 1}$$
$$\frac{f(4) - 10}{3} \geq 2$$
$$\boxed{f(4) \geq 16}$$

Problem 10

Does there exist a function f such that $f(0) = -1, f(2) = 4$, and $f'(x) \leq 2$ for all x?

$$f'(c) = \frac{f(2) - f(0)}{2 - 0} = \frac{5}{2}$$
$$f'(c) > 2$$
$$\boxed{f(x)\,\text{DNE}}$$

Problem 11

Let $f(x) = \dfrac{1}{x}$ and $g(x) = \begin{cases} \dfrac{1}{x} & x > 0 \\ 1 + \dfrac{1}{x} & x < 0 \end{cases}$

Show that $f'(x) = g'(x)$ for all x in their domains. Can we conclude from the Corollary that $f - g$ is constant?

$$f'(x) = -\frac{1}{x^2}$$
$$g'(x) = \begin{cases} -\dfrac{1}{x^2} & x > 0 \\ 0 - \dfrac{1}{x^2} & x < 0 \end{cases}$$
$$\boxed{f'(x) = g'(x), \quad \forall x \in (-\infty, 0) \cup (0, \infty)}$$

No. The domain is not one single interval.

Problem 12

Two runners start a race at the same time and finish in a tie. Prove that at some time during the race, they have the same speed. [Hint: Consider $f(t) = g(t) - h(t)$, where g and h are the position functions of the two runners.]

$$f(0) = g(0) - h(0) = 0$$
$$f(t) = g(t) - h(t) = 0$$
$$f(0) = f(t) \implies f'(c) = 0$$
$$g'(t) - h'(t) = 0$$
$$\boxed{g'(t) = h'(t)}$$